Experiment No. 13: Affine Transformation of an Image

Aim: To perform affine transformation on a given image using Python and OpenCV.

Program (Google Colab):

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload an image
uploaded = files.upload()
image_path = next(iter(uploaded))

# Read the image
img = cv2.imread(image_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

h, w = img.shape[:2]

# Define three points in the original image
pts1 = np.float32([[50, 50], [200, 50], [50, 200]])

# Define corresponding points in the transformed image
pts2 = np.float32([[10, 100], [200, 50], [100, 250]])

# Compute affine transformation matrix
M = cv2.getAffineTransform(pts1, pts2)
```

```
# Apply affine transformation
transformed = cv2.warpAffine(img, M, (w, h))

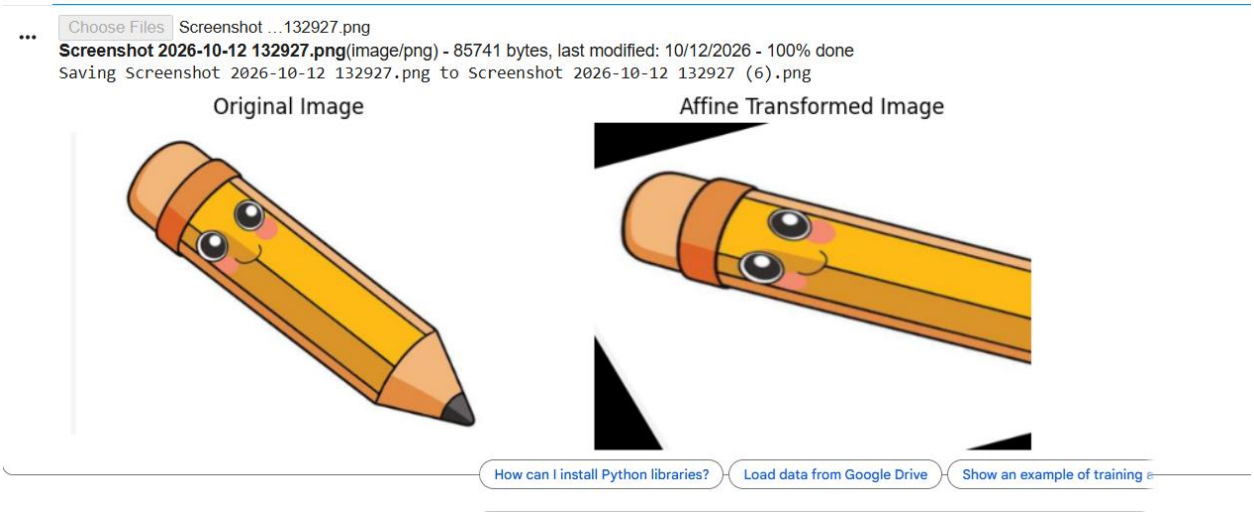
# Display original and transformed images
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(transformed)
plt.title("Affine Transformed Image")
plt.axis("off")

plt.show()
```

OUTPUT:



Expected Output:

1. Original image.
2. Affine-transformed image showing changes in position, shape, or orientation.

Result: Affine transformation was successfully performed on the given image using Python and OpenCV. The transformation was applied using cv2.getAffineTransform() and cv2.warpAffine().